

intrinsic π -phase inversion of the echo pulse suppresses these coherence-induced oscillations. As a result, the spectral response remains smooth and well-behaved, enabling reliable linewidth extraction in regimes where standard and Lorentzian pulses display pronounced oscillatory distortions; see the additional high-amplitude data in the Supplemental Material [18].

Second, the use of formally infinite-support pulses necessitates the introduction of a cutoff parameter, as discussed previously. Consequently, the relevant parameter space is effectively three-dimensional: pulse duration, drive amplitude, and cutoff level.

Motivated by the stability observed at fixed pulse duration, we restrict the analysis to constant-length pulses and examine the remaining two parameters. In Fig. 4 we simulate both the root-Lorentzian pulse and its echo-modified counterpart. We extract the full width at half maximum (FWHM) via fitting (as in Fig. 3) and evaluate the normalized metric given by the linewidth divided by the peak signal amplitude. This quantity is strongly correlated with spectral resolution and is directly connected to the inverse Fisher information, i.e., the Cramér–Rao bound.

The results reveal that the root Lorentzian pulse achieves narrow linewidths across a relatively large region of parameter space. However, its performance is not uniform and exhibits oscillatory structure for the chosen pulse duration. In contrast, the echo-root Lorentzian pulse produces even narrower linewidths while maintaining a significantly broader and more uniform high-resolution region. Although the optimal region may appear small at first glance, the logarithmic scale of the parameter axes should be noted. More importantly, the high-resolution region forms a robust and contiguous “island,” indicating strong parameter insensitivity. This robustness makes the echo protocol substantially more stable against parameter variations and therefore highly advantageous for practical spectroscopy.

CONCLUSION

In conclusion, we have demonstrated that the echo–Lorentzian pulse constitutes a promising and practical approach for high-resolution spectroscopy. By combining adiabatic shaping with an intrinsic echo phase inversion, the protocol significantly enhances robustness against drive-amplitude variations. Remarkably, it broadens the operational amplitude window by approximately three orders of magnitude, while maintaining high spectral resolution even at drive strengths approaching neighboring qubit transitions. These properties make the echo–Lorentzian protocol particularly attractive for precision qubit characterization under realistic experimental constraints.

This opens the route to coherent spectroscopy based on

phase-sensitive off-resonant response rather than population transfer.

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